

EDUCATION

University of Mons (UMONS)

Mons, Belgium

Ph.D. in Software Engineering, Advisor: Prof. Tom Mens

2021, Oct, 21 – 2026, May, 06

– Thesis title: “Understanding the Evolution of GitHub Actions Automation Workflows”

Sharif University of Technology (SUT)

Tehran, Iran

M.S. in Software Engineering, Advisor: Dr. Abbas Heydarnoori, GPA: 18.30/20

2019, Sept, 01 – 2021, Sept, 20

– Thesis title: “Automatic Linking of Issue Reports and Commits in Software Repositories”

K. N. Toosi University of Technology (KNTU)

Tehran, Iran

B.S. in Computer Engineering, Advisor: Dr. Ali Ahmadi, GPA: 17.64/20

2015, Sept, 01 – 2019, Jun, 20

– Thesis title: “Deaf Disease Genes Detection Using Artificial Intelligence Tools”

RESEARCH EXPERIENCES

• Research

- **2026, May, 15 - Current:** PostDoc Researcher @ DistriNet Group T @ KU Leuven
- **2021, Oct, 21 - 2026, May, 06:** PhD Researcher @ SEL Lab @ University of Mons
- **2019, Oct, 01 - 2021, Sept, 20:** Master Student Researcher @ ISE Lab @ SUT
- **2019, Jan, 01 - 2019, Aug, 01:** Research Internship @ IIPL @ KNTU

• Mobility

- **2022, Oct, 17 - 2023, Jan, 13:** FNRS-funded Research Stay, Visiting Researcher @ Radboud University

• Organizations

- **2022, Apr, 10 - 2022, Sept, 21:** Workshop publicity chair for BENEVOL 2022

• Academic Service

- Participating in **journal peer review** for:
 - IEEE Software Journal (**IEEE Softw.**)
 - Journal of Systems and Software (**JSS**)
 - Empirical Software Engineering (**EMSE**)
 - Science of computer programming (**SCICO**)
 - Transactions on Software Engineering (**TSE**)
- Program Committee:
 - ICPC 2027
 - SQA4AI 2027
 - MSR 2027
 - TechDebt 2027
 - TechDebt 2026 (Junior)
 - BENEVOL 2026
 - ESEM-ESEIW 2026

• Talks

- **2023, Mar:** Presented qualitative research methodology in the form of a full-session lecture to master’s students

TEACHING EXPERIENCES

- **Teaching Assistant**
 - **2026, Sept, 01 - 2026, Dec, 10: Object Oriented Programming** To Software Engineering **bachelor** students @ Faculty of Technology @ KU Leuven - Group T
 - **Feb, 02 - May, 24 of 2022, 2023, 2024, 2026: Software Evolution** To Computer Science **master** students @ Faculty of Sciences @ UMONS
 - **2020, Sept, 01 - 2020, Dec, 10: Data Science for Software Engineering** To Software Engineering **master** students @ Faculty of Sciences @ SUT
 - **2019, Jan, 10 - 2019, May, 20: Database Design** To Software Engineering **bachelor** students @ Faculty of Sciences @ SUT
 - **2018, Sept, 01 - 2018, Dec, 10: Data Structure & Database Design** (two separate courses) To Software Engineering **bachelor** students @ Faculty of Sciences @ KNTU
- **Master Students Defense**
 - **2024, Jun:** Attend as a jury member of **master** thesis defense @ Faculty of Science @ UMONS

PROFESSIONAL EXPERIENCES

- **2025, Jan, 05 - 2026, Jan, 05:** Cloud Engineer/Data Analyst @ Necko Technologies

SELECTED PUBLICATIONS (GOOGLE SCHOLAR LINK)

1. **P. Rostami Mazrae**, A. Decan, T. Mens, M. Wessel, “An Empirical Study of the Evolution of GitHub Actions Workflows”, *Journal of Systems and Software (JSS)*, 2026.
<https://doi.org/10.1016/j.jss.2026.112824>
2. M. Wessel, T. Mens, A. Decan, **P. Rostami Mazrae**, “The GitHub Development Workflow Automation Ecosystems”, *Software Ecosystems: Tooling and Analytics*, Book chapter, Springer Nature, 2023.
https://doi.org/10.1007/978-3-031-36060-2_8
3. **P. Rostami Mazrae**, T. Mens, A. Decan, M. Golzadeh, “On the usage, co-usage and migration of CI/CD tools: a qualitative analysis”, *Empirical Software Engineering Journal (EMSE)*, 2023.
<https://doi.org/10.1007/s10664-022-10285-5>
4. A. Decan, T. Mens, **P. Rostami Mazrae**, M. Golzadeh, “On the Use of GitHub Actions in Software Development Repositories”, *International Conference on Software Maintenance and Evolution (ICSME)*, 2022.
<https://doi.org/10.1109/ICSME55016.2022.00029>
5. **P. Rostami Mazrae**, M. Izadi and A. Heydarnoori, “Automated Recovery of Issue-Commit Links Leveraging Both Textual and Non-textual Data”, *International Conference on Software Maintenance and Evolution (ICSME)*, 2021.
<https://doi.org/10.1109/ICSME52107.2021.00030>

OPEN SOURCE TOOLS

- **gawd (GitHub Actions Workflow Differ) – Python** 2024
Developed an open-source Python library and CLI tool to compute structural differences between GitHub Actions workflows, identifying added, removed, moved, renamed, or modified items with syntax-aware comparison.
Link: <https://github.com/pooya-rostami/gawd>
Publication: <https://doi.org/10.1145/3643991.3644873>

SCHOLARSHIPS AND AWARDS

- **FNRS mobility funding inside Europe**, 2022, Total of 5,950 euros, Radboud University - The Netherlands
- **ICSME 2022 Distinguished Paper Award** for “On the Use of GitHub Actions in Software Development Repositories”. (<https://cyprusconferences.org/icsme2022/>)
- **MSR 2022 best Hackathon Paper Award** for “Bot detection in GitHub repositories”. (<https://conf.researchr.org/info/msr-2022/awards>)

CERTIFICATE

- **“AWS Certified Solutions Architect – Associate”** May, 2025
Certificate link: <https://www.credly.com/badges/0db7e80d-1a63-45da-9501-fe17a607f6cf>
- **“AWS Certified Cloud Practitioner”** Feb, 2025
Certificate link: <https://www.credly.com/badges/42d89f68-b12b-4b50-b38c-3f5fb79423bf>
- **“The Power of Statistics” Coursera certificate** Sep, 2023
Certificate link: <https://coursera.org/share/c0bad269d416082d0511e640cdc588b0>

TECHNICAL SKILLS

- **Software Development Technologies & Methodologies:** Agile (Scrum), DevOps, IDEs (VS Code, IntelliJ, JupyterLab), testing framework (pytest), Git, Atlassian (Bitbucket, Jira), Docker, Linux/Unix
- **Coding:** Python, TypeScript, HTML, CSS
- **Database:** RDBMS such as MySQL, MS SQL
- **Data Analysis Libraries & Technologies:** Pandas, Keras, sci-kit learn, NumPy, NLTK, HuggingFace, PySpark
- **ML Algorithms:** Classical Machine learning, and deep learning algorithms (SVM, Random Forest, CNN, LLMs)
- **AWS:** S3, Glue, Athena, Lambda, Step Functions, RDS, Kinesis, QuickSight, etc.

LANGUAGES

English: Fluent

French: Intermediate Proficiency (B1 ongoing)

Persian: Native